

Emily Miller

 emilymiller.xyz  LinkedIn  python151

Education

University of Edinburgh

BSc Computer Science and Mathematics

Sep 2024 - May 2028

- Predicted First Class Honors
- Coursework: Computer Security, Computer Architecture and Design, Computer Networks

Professional Experience

Interrupt Labs

Vulnerability Research Intern

Basingstoke, UK
Jul 2025 - Aug 2025

- Completed intensive in-house training led by senior security researchers.
- Conducted static and dynamic analysis and built internal tooling.
- Discovered several critical vulnerabilities in target device and developed proof-of-concept exploits.

Textron

Cloud Automation Intern

Hurst, TX
Sep 2023 - Sep 2024

- Developed enterprise firewall change request process to streamline firewall changes in ServiceNow. Integrated with 7 business units and 3 vendors, impacting 350+ tickets.
- Identified exploitable weaknesses in internal deployment and infrastructure management systems that could have enabled destructive actions across organizational infrastructure.
- Integrated with the Azure Service Bus REST API to automate implementation of operational readiness reviews (ORRs).
- Improved various operational processes through automation including assisting in developing our VM modification and decommission processes by implementing, testing, and pushing 17 change requests through to production.

Cyber Security Engineering Intern

Aug 2022 - Aug 2023

- Managed our Splunk SIEM instance, working to deploy Risk Based Alerting (Splunk RBA) into enterprise environment. Worked closely with stakeholders in Cyber Incident Response Team (CIRT) and Security Operations Center (SOC) during development and deployment.
- Engaged SOC to develop new endpoint and server security controls using Group Policy in response to Red Team exercises.
- Applied Databricks and Python to develop an ETL pipeline between our Azure Data Lake and Splunk, optimizing data integration and analysis.

Other Experience

SIGINT: Cyber Security Society

President

University of Edinburgh
May 2025 - May 2026

- Led organisation of pwnEd 7, SIGINT's flagship CTF and conference, drawing 100+ participants nationwide and hosting 8 industry and academic speakers.
- Built Infrastructure for the Beginner CTF and Mini CTF, handling over 100 concurrent users with no downtime.
- Built CTF Challenges for 5 CTFs for the Binary Exploitation, Cryptography, and Misc categories.
- Deployed on-premises infrastructure in a colocation (co-lo) environment using Proxmox as the hypervisor, and built LDAP-based IAM alongside Infrastructure-as-Code (IaC) backed core services using NixOS.
- Designed and implemented CI/CD pipelines for legacy infrastructure, automating previously manual deployments and enabling streamlined website and service configuration updates.

CompSoc

Technical Secretary

University of Edinburgh
May 2025 - May 2026

- Deployed and managed distributed vLLM inference infrastructure across 48 GPUs on EIDF compute clusters for an AI-focused hackathon, handling Kubernetes orchestration, resource allocation, and participant access management.
- Migrated legacy infrastructure to a cloud-native managed Kubernetes cluster, building CI/CD pipelines for 7 services and automating external backup processes.

Write-ups

Kernel Exploitation Infrastructure 

Oct 2025

CTF Hosting Guide 

Aug 2025

Technologies

Technologies: Binary Ninja, GDB, C, Python, x86 Assembly, ARM Assembly, Frida, AFL++, pwntools, Embedded Linux